

Lab 6: Metamorphic Rocks Prelab Reading

Classifying Metamorphic Rocks

Metamorphism relates to changes in the mineral composition and/or texture that occur in solid rocks as a result of increasing pressure and/or temperature and reactions with hot, mineral-rich fluids. These conditions will result in three possible changes in the rocks:

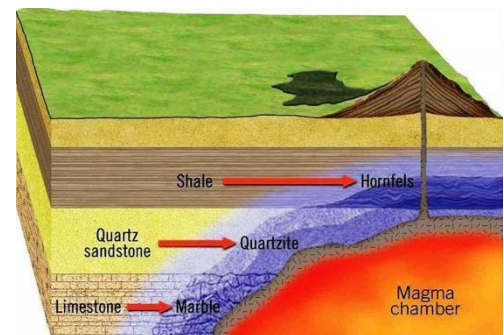
- Mineral changes - some minerals may simply recrystallize to larger grains; some minerals may be converted to more stable minerals.
- Physical changes - High pressures often result in bending or folding of rocks and distortion of mineral grains.
- Fluids - new minerals may be precipitated from hot fluids to fill fractures in the rock and form veins.



There are two main types of metamorphism: contact metamorphism and regional metamorphism. Metamorphic rocks require a temperature range that is high enough to trigger the chemical reactions necessary to change minerals but not so high that the minerals will melt. Temperatures of at least 200° C are generally required before some of the minerals in a rock will undergo chemical reactions to cause compositional changes. Depending upon their composition, most minerals in rocks near Earth's surface melt at temperatures ranging from 600-1100° C. Such temperatures are common near magma chambers or at deeper levels in the crust.

Contact Metamorphism

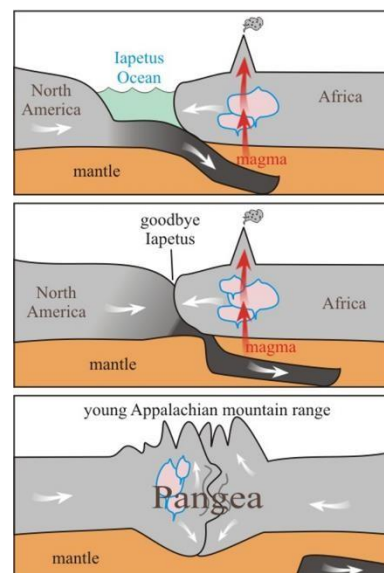
Contact metamorphism occurs when rocks undergo metamorphism because they come in contact with a heat source (usually a magma body). A comparable change occurs in roasting meat or baking bread - the composition does not change but the texture of the material does. Marble is an example of a rock that may be formed by contact metamorphism. Marble forms when limestone is heated to high temperatures. Both marble and limestone may have the same composition, but marble typically has larger grains. The dark rock, hornfels, forms when fine grained clastic rocks like mudstone (shale) are in contact with a heat source.

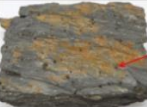


Regional Metamorphism

Regional metamorphism occurs when rocks undergo increased temperatures and pressures typically associated with the plate tectonic processes that form mountains. In these areas rocks may be buried to great depths. The additional pressure causes sheet-like minerals (e.g., mica) in the rock to physically rotate or to grow with a preferred alignment. Higher temperatures, pressures and fluids cause some types of minerals to grow. Those minerals are layered parallel to each other (like a stack of papers on a table) and perpendicular to the direction of pressure (like pressing down on the stack of papers). This geometric orientation generates an alignment of minerals into sheets

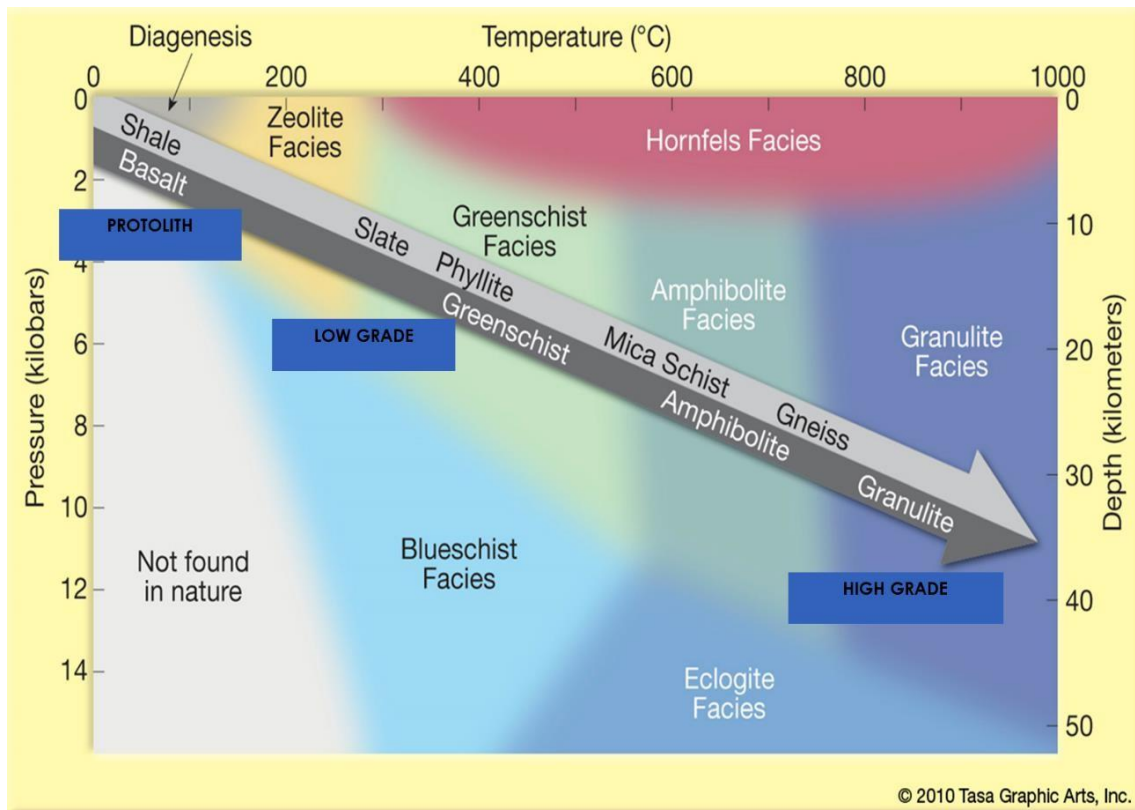
termed a foliation. Foliations may also be represented by sheets of flattened objects or bands of different colored grains. The higher the temperatures and/or pressures the more intense the metamorphism. Many rocks formed by regional metamorphism can be classified by the size of minerals that form (a temperature and pressure dependent process) when they change from the original rock to the metamorphic rock (see table). Foliated metamorphic rocks, in order of increasing metamorphism are slate, phyllite, schist, and/or gneiss. Grain size in slate is much too small to be seen with the naked eye and the presence of millions of tiny crystals of sheet-like minerals means that slate splits easily into thin sheets. Mica is a common mineral in these rocks and can give some of the rocks (e.g., phyllite, schist) a shiny, silky appearance often associated with a wavy foliation. Gneiss (pronounced “nice”) has the largest grain size with light and dark colored minerals typically separating into lighter and darker bands in the rock. Not all rocks that undergo regional metamorphism will form a foliation. Rocks that lack sheet-like minerals will not create a foliation. Sandstone, composed mainly of quartz grains, will be converted to quartzite by regional metamorphism; limestone will form marble. Neither metamorphic rock contains foliations.



TEXTURE	CHARACTERISTICS	PROTOLITH	METAMORPHIC ROCK
NON FOLIATED	Blocky grains of quartz (hardness 7).	Sandstone or Siltstone	Quartzite 
	Blocky grains of calcite (hardness 3). Fizzes with dilute HCl.	Limestone	Marble 
	Fine-grained, various colours.	Shale or Basalt	Hornfels 
	Soft (hardness ~3), glossy, and black, with low specific gravity (~1.4).	Bituminous coal	Anthracite 
FOLIATED	Alternating bands of light- and dark-coloured minerals (dark minerals are usually biotite or amphibole), called gneissic banding.	Shale or Igneous Rock	Gneiss 
	Contains shiny muscovite (light) or biotite (dark) micas. May also have quartz, talc, garnet (red arrow), amphibole. Schistose pattern of foliation.	Shale	Schist 
	Fine-grained rock with the grains only visible as a satin sheen. Similar to slate, but with a satin lustre and may have wrinkled cleavage.	Shale	Phyllite 
	Very fine-grained rock, tends to split in parallel fragments (known as slaty cleavage).	Shale	Slate 

Metamorphic Facies

Metamorphic rocks form under a variety of conditions within Earth's crust. These varying temperature and pressure conditions are known as facies and are characterized by different assemblages of minerals. Identifying a metamorphic rock tells you about the conditions under which it formed. Look at the chart below. Shale follows a path in which it first is metamorphosed into slate. As pressure and temperature increase, it becomes phyllite then mica schist and finally gneiss. So, if you find a mica schist, you this formed at approximately 25km depth at 600°C.



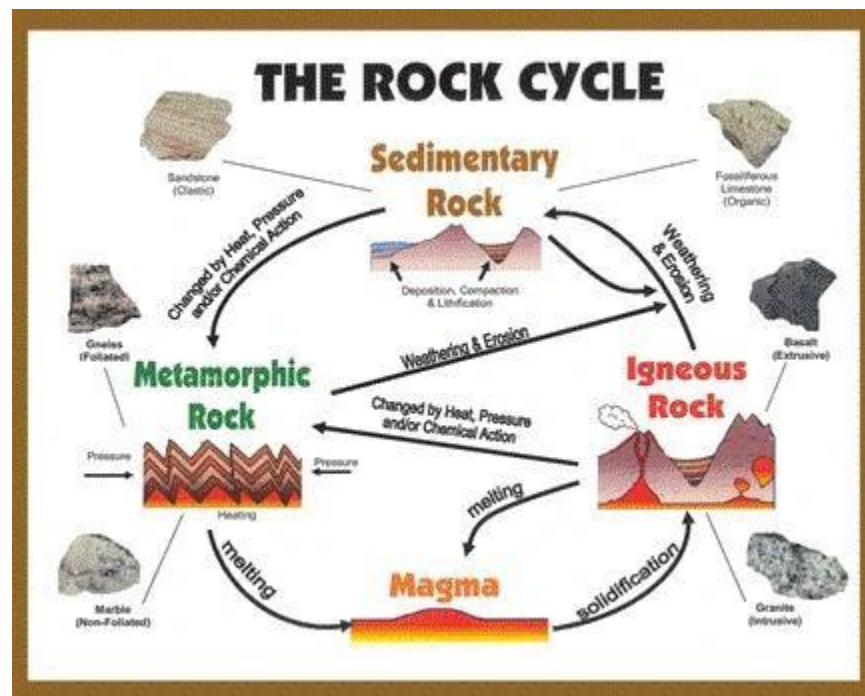
Rock Review

Igneous rocks form as magma cools. Extrusive igneous rocks cool at the Earth's surface. Since temperatures are much cooler than the melting point of the magma, they cool quickly which limits crystal growth. This creates a fine grain (aphanitic) texture. If they cool extremely quickly, they can form a glassy texture in which no crystals are present (i.e. obsidian). Intrusive igneous rocks cool beneath the surface. Since temperatures are not much cooler than that of the magma, they cool slowly, allowing for larger crystals to grow. This creates a coarse grained (phaneritic) texture. Igneous rocks are also divided by composition. Felsic igneous rocks contain high amounts of silicate minerals and are usually a lighter color because of these minerals. Mafic igneous rocks contain high amounts of iron and magnesium which make them darker in color and denser.

Sedimentary rocks form from the breakdown, transport, deposition and lithification of other rocks. Sedimentary rocks tell us a lot about past environments and often contain fossils. There are three

main classifications of sedimentary rocks: clastic, chemical, and biochemical. Clastic form from the compaction and cementation (lithification) of small rock particles. Chemical sedimentary rocks form from minerals crystallizing out of solution (usually water) in which minerals were dissolved during chemical weathering processes. Bioclastic sedimentary rocks form from the lithification of organic material such as shells or plant matter (coal).

Metamorphic rocks form when rocks experience an increase in pressure and /or temperature that forces the minerals to recrystallize to a more stable mineral. There are two main types of metamorphism, contact and regional. Contact metamorphism occurs when magma heat rocks around it increasing the temperature without melting the rock. Regional metamorphism occurs when a large amount pressure is applied to rocks, usually at convergent plate boundaries where plates are colliding. Metamorphic rocks are classified by their texture. Foliated metamorphic rocks have parallel bands of minerals caused by the aligning of crystals in the same direction from extreme pressure. Non-foliated metamorphic rocks do not have these parallel bends and usually form during contact metamorphism from extreme heat but can form from extreme pressure as well.



If you want to watch a really cheesy 80s style video on the rocks cycle, check out the “[Rock Odyssey](#)”. Warning though, you will never get the songs out of your head.